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Channel Estimation for mmWave mMIMO Systems with Low-Resolution ADCs Based on Adaptive Quantization Improved CGAN

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***Abstract*—Millimeter wave massive multiple-input multiple-output systems offer higher information capacity and can mitigate severe propagation losses. However, the rapid increase in the number of antennas results in higher hardware costs and power consumption. Although these issues can be addressed by reducing the resolution of analog-to-digital converters (ADCs) in the radio frequency chain, low-resolution ADCs introduce significant nonlinear errors when quantizing received signals, making accurate channel estimation very challenging. To mitigate the impact of quantization noise from low-resolution ADCs on channel estimation performance, we propose a novel channel estimation method based on adaptive quantization conditional generative adversarial network (AQ-CGAN). Instead of using traditional quantization methods, we convert the quantized received signals and pilot sequences as well as the quantization thresholds into two-dimensional images, randomly generate and iteratively optimize the thresholds under the CGAN framework. At the same time, we use the neural network and CGAN to establish a mapping relationship between quantization thresholds and channel matrices to achieve adaptive quantization, and introduce a new loss function to improve the channel estimation accuracy in massive MIMO systems. Simulation results demonstrate that the proposed algorithm outperforms traditional quantized CGAN channel estimation under the same signal-to-noise ratio conditions.**

***Index Terms*—Low-resolution, analog-to-digital converter, channel estimation, massive MIMO, conditional generative adversarial network**

I. INTRODUTION

THe combination of millimeter wave (mmWave) and massive multiple input multiple output (mMIMO) system meets the requirements of fifth generation (5G) mobile communications for higher system capacity and performance. By deploying large-scale antenna arrays on access point or base station (BS), the utilization of bandwidth resource can be significantly improved, information capacity can be increased, anti-interference ability can be enhanced, and data transmission rate can be significantly improved, while the BS density and system bandwidth remain unchanged [1],[2]. However, massive MIMO systems equipped with high-resolution analog-to-digital converters (ADCs) result in an exponential increase in power consumption and hardware complexity, resulting in huge hardware costs and excessive energy consumption [3]-[5].

The use of low-resolution ADCs, such as those with 1-3 bits, instead of traditional high-precision ADCs is one of the main approaches to solve this problem. Significant previous work has been done on millimeter wave massive MIMO systems with low-resolution ADCs. Approximate closed-form expressions and channel capacity analysis for the achievable rate of the uplink MIMO system in [6]-[8] are presented, and it is shown in [8] that the effect of low-resolution ADCs on the system can be compensated by increasing the number of antennas. In [9], the spectral efficiency loss using a low-resolution ADC is only 20% compared to an unquantized signal with moderate SNR. Compared with the traditional high resolution ADC, the performance loss of low resolution ADC in massive MIMO system is only 1.96dB at high signal-to-noise ratio (SNR) [10], which provides the possibility for practical application. Although the use of low-resolution ADCs reduces power consumption and cost, it introduces large quantization errors that make channel estimation more difficult and complex.

Firstly, in order to deal with the challenge of dealing with massive data in channel estimation, several algorithms have been developed in [11]-[15]. The authors of [11] proposed a distributed compressed sensing (CS)-based channel estimation scheme for millimeter-wave MIMO systems, which employs an adaptive measurement matrix and a grid-matching tracking strategy, and its channel estimation performance outperforms that of conventional CS algorithms. The approximate message passing algorithm (AMP) proposed in [12] is based on Bayesian framework, which has fast convergence and low complexity. In [13], orthogonal matching Pursuit algorithm (OMP) is applied. This greedy iterative employs an over-complete dictionary to approximate the original signal. The iterative hard threshold algorithm (IHT) was proposed in [14] using a fast gradient descent algorithm with constant step size and hard thresholding to solve the problem iteratively. [15] first proposed a quantum computing architecture to optimize the channel estimation for millimeter wave mMIMO systems. Quantum computing is used to handle the large-scale matrix operations brought by a large number of antennas.

In order to mitigate the quantization noise caused by low-resolution ADCs, an estimator proposed in [16] combines

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orthogonal matching pursuit to reduce the quantization noise. Reference [17] applies Bussgang decomposition to transform the nonlinear quantization error into an additive quantization noise for performance analysis. While this method facilitates analysis, it does not effectively reduce the impact of quantization noise. In [18], the quantized iterative hard thresholding (QIHT) algorithm is introduced, incorporating a quantization model into the traditional iterative hard thresholding (IHT) channel estimation method to address the nonlinearity during the quantization process. The performance of the QIHT algorithm is analyzed under different SNRs and ADC precisions, demonstrating advantages such as low complexity and high robustness. However, it exhibits slow convergence speed. Therefore, the aforementioned traditional algorithms perform poorly when dealing with large data volumes and quantization errors caused by low-resolution ADCs. For example, these algorithms typically require long training sequences, consuming a large amount of pilot resources and reducing spectral efficiency. Additionally, the system requires a large number of iterations to converge, leading to increased computational complexity and latency issues. When the system operates under low SNR conditions, performance degrades significantly. Furthermore, some methods are based on assumptions such as ideal hardware and high power consumption, which cannot be achieved in practical applications. Additionally, millimeter-wave channels exhibit sparsity and high temporal variability, yet traditional quantization methods often overlook these sparse characteristics and struggle to adapt to such dynamic changes, potentially rendering real-time updates unfeasible in high-speed mobile scenarios.

Recently, [19]-[23] apply DL to channel estimation in mMIMO systems with low-resolution ADCs. In [19], it is shown that trained single neural network is more robust in dealing with ADC impairments compared to a variant of generalized approximate message passing (GAMP). In [20], it was demonstrated that utilizing prior channel estimation observations and deep neural networks (DNNs) can learn the non-trivial mapping from quantized received measurements to the channel. Reference [21] proposed a two-stage estimation scheme for channel estimation in 1-bit ADC mMIMO systems using DNNs, which achieved higher gains compared to the baseline linear minimum mean square error (LMMSE) estimator. For the uplink of 1-bit mMIMO systems, a channel estimation method using DNNs on multiple signal segments was introduced in [22]. Additionally, a channel estimation method for low-resolution mMIMO systems was proposed in [23], where DNNs were utilized as autoencoders to optimize training signals and combined with minimum mean square error (MMSE) channel estimation. These studies focus on 1-bit or few-bit resolution scenarios, leaving room for further improvement in channel estimation accuracy.

In order to further improve estimation accuracy, a two-stage channel estimation method was proposed in [24]. This method involves initially making a rough estimation of the received signals, followed by the use of a neural network to enhance the estimation accuracy. Additionally, [25] introduced a prior-assisted Gaussian mixture-approximated LAMP (GM-LAMP) method for beam-space channel estimation. By utilizing prior information modeled by a Gaussian Mixture distribution, they replaced the original contraction function in the LAMP network with a differentiable Gaussian Mixture contraction function, innovatively proposing the prior-assisted GM-LAMP network to achieve more accurate channel estimation. Furthermore, [26] utilized Conditional Generative Adversarial Networks (CGAN) to predict the real channel by adversarially training two deep learning networks. This network not only learned the mapping from quantized observed real channels, but also learned an adaptive loss function to properly train the network.

However, there is still significant scope for further development in improving the accuracy of channel estimation in these studies. Previous deep learning approaches have focused on learning the mapping from quantized signals to channel matrices. In order to mitigate the impact of quantization errors on channel estimation, [27] developed an adaptive quantization (AQ) scheme, where the quantization threshold dynamically adjusts to converge to the optimal threshold. By identifying the optimal quantization threshold, the impact of quantization errors on channel estimation can be reduced. While most studies assume a fixed quantization threshold, they may deviate significantly from the optimal quantization threshold. The theoretical analysis in [28] indicates that the optimal quantization threshold is a function of unknown channel parameters, and with optimal threshold design, using low-resolution ADCs can achieve estimation errors close to those with infinite precision ADCs. Our work can be considered an extension of previous research on adaptive quantization design, specifically applicable to channel estimation with low-resolution ADCs. To enhance denoising generalization capabilities and estimation accuracy, we introduce quantization thresholds into DL to improve Conditional Generative Adversarial Networks (CGANs).

This paper proposes CGAN-based channel estimation methods for MIMO systems employing low-resolution ADCs. The main contributions are summarized as follows:

- This paper proposes an Adaptive Quantization Conditional Generative Adversarial Network (AQ-CGAN) algorithm, which transforms quantized received signals and pilot sequences along with quantization thresholds into two-dimensional images. Utilizing CGAN as the underlying framework, a set of distinct thresholds is stochastically generated based on certain statistical prior knowledge of the channel. Subsequently, these quantization thresholds are employed as conditional information to enhance the input of CGAN. Through the discriminator within CGAN, the quantization thresholds are discriminated and refined iteratively to approximate the optimal thresholds. This process improves the accuracy of channel estimation in massive MIMO systems employing low-resolution ADCs.
- This paper incorporates quantization thresholds as one of the inputs to the neural network and establishes a mapping relationship between quantization thresholds and estimated channel matrices using the CGAN network, enabling adaptive quantization. Additionally, the introduction of a new loss function can improve the optimization direction of the neural network, ensuring

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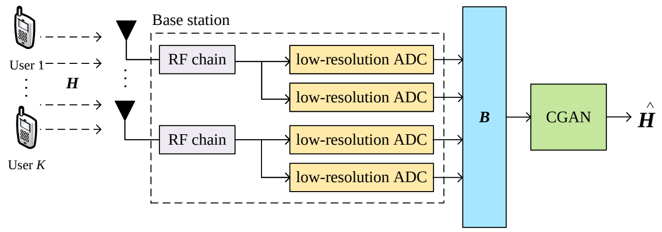

Fig. 1. Low-Resolution ADCs mmWave mMIMO Systems channel estimation framework.

good channel estimation performance even in low signal-to-noise ratio environments.

- Simulation analysis demonstrates that the proposed AQ-CGAN algorithm exhibits superior channel estimation performance compared to conventional quantization-based channel estimation methods across 1-bit, 2-bit, and 3-bit quantization scenarios. Specifically, under conditions of 1-bit ADC and a signal-to-noise ratio of 25dB, AQ-CGAN achieves an approximate 0.28% enhancement in channel estimation accuracy compared to CGAN.

The paper is arranged as follows. Section II introduces the signal model of the mmWave mMIMO system using low-resolution ADCs. In Section III, we elaborate the design of the proposed method, namely the AQ-CGAN-based channel estimator. In addition, the simulation analysis in Section IV is used to evaluate the performance of the proposed method. Finally, Section V presents the conclusions of this paper.

II. SYSTEMS MODEL

We consider a multi-user millimeter-wave massive MIMO system based on low-resolution ADCs, and Fig. 1 shows its channel estimation framework. In this scenario, there are K user terminals, each equipped with one antenna, while the base station (BS) is equipped with M antennas and each BS antenna is connected to two low-resolution ADCs [29]. We use the DeepMIMO framework to generate channel datasets between the BS and multiple users, which are further based on accurate ray tracing data simulated by InSite, a wireless ray tracing simulator developed by Remcom [30]. To characterize the finite scattering characteristics, there are scatterers L_k between user k and the BS. Assuming each scatterer contributes to a single propagation path between the base station and the user, the channel in a single-antenna user system is modeled as

$$\boldsymbol{h}_k = \sqrt{\frac{M}{L_k}} \sum_{l=1}^{L_k} \boldsymbol{\omega_l} \mathbf{a}\left(\theta_l^k\right), \tag{1}$$

where L_k is the number of paths and $\omega_l \sim \mathcal{CN}\left(0, \sigma_\omega^2\right)$ is the propagation gain of the l-th path with σ_ω^2 being the average power gain. For the l-th path, θ_l^k is the azimuth angles of arrival (AoA) at the BS and $\mathbf{a}\left(\theta_l^k\right)$ denotes the corresponding response vector. For a uniform linear array, ω_l and $\mathbf{a}\left(\theta_l^k\right)$ are given by

$$\omega_l = \sqrt{\frac{P_l^k}{K}} \mathrm{e}^{\mathrm{j}\left(\phi_l^k + \frac{2\pi k}{K}\lambda_l^k B\right)} \tag{2}$$

$$\mathbf{a}\left(\theta_l^k\right) = \frac{1}{\sqrt{M}}\left[1, e^{-j2\pi\frac{d}{\lambda}\sin\left(\theta_l^k\right)}, \ldots, e^{-j2\pi\frac{d}{\lambda}(M-1)\sin\left(\theta_l^k\right)}\right]^T, \tag{3}$$

where d denotes the space between the adjacent antennas at the BS and λ is the wavelength of the carrier frequency. In the channel path l between the base station and the k-th user, ϕ_l^k represents the phase, P_l^k denotes the received power, and λ_l^k represents the transmission delay. B is the system bandwidth.

The full channel matrix $\boldsymbol{H} \in \mathbb{C}^{M \times K}$ between the BS and the K users can be represented as

$$\boldsymbol{H} = \left[\boldsymbol{h}_1, \boldsymbol{h}_2, \cdots, \boldsymbol{h}_k, \cdots, \boldsymbol{h}_K\right] \tag{4}$$

As illustrated in Fig. 1, we perform channel estimation at the BS by utilizing pilot signals from users. K users simultaneously transmit pilot sequences of length v to the BS. Thus, the unquantized received signal $\boldsymbol{Y} \in C^{M \times v}$ at the BS can be expressed as

$$\boldsymbol{Y} = \boldsymbol{H\Phi} + \boldsymbol{W} \tag{5}$$

where $\boldsymbol{\Phi} \in \mathbb{C}^{K \times v}$ represents the pilot matrix composed of pilot sequences sent by K users, each user's pilots in Φ is mutual orthogonal. The noise vector $\boldsymbol{W} \in \mathbb{C}^{M \times v}$ is modeled as a circularly symmetric complex Gaussian random variable with zero mean and variance $2\sigma_w^2$. At each antenna, the real and imaginary parts of the received signal are quantized using a pair of low-resolution ADCs. Therefore, a total of $2M$ low-resolution ADCs are required. The quantized output of the received signal $\boldsymbol{B} \in \mathbb{C}^{M \times v}$ can be expressed as

$$\begin{aligned} \boldsymbol{B} &= \mathcal{Q}_b\left(\boldsymbol{Y}\right) \\ &= \mathcal{Q}_b\left(\Re\left\{\boldsymbol{Y}\right\}\right) + j\mathcal{Q}_b\left(\Im\left\{\boldsymbol{Y}\right\}\right), \end{aligned} \tag{6}$$

where $\mathcal{Q}_b(.)$ is a function representing the quantization operation on each real and imaginary components. b represents the number of bits for quantization by low-resolution ADCs. For example, the signum function $\mathrm{sgn}(\cdot)$ is an element-wise operator for one bit quantization defined as

$$\mathrm{sgn}(x) = \begin{cases} 1, x \geq 0, \\ -1, \text{ otherwise}, \end{cases} \tag{7}$$

It is applied to both the real and imaginary part of the argument x. Thus, $\boldsymbol{B}$ is a quantized signal where its element takes values from the set $\{1+j, 1-j, -1+j, -1-j\}$.

Note that in (5), we implicitly assume the quantization threshold to be zero. However, using the same zero threshold for all measurements may not be optimal, and analyzing the impact of quantization thresholds on channel estimation performance is highly significant. Let $\boldsymbol{T} \in \mathbb{C}^{M \times v}$ denote the threshold used for quantization with low-resolution ADCs. The quantized received signal $\boldsymbol{B}$ can be further expressed as

$$\boldsymbol{B} = \mathcal{Q}_b(\boldsymbol{H\Phi} + \boldsymbol{W} - \boldsymbol{T}). \tag{8}$$

The ultimate goal of channel estimation is to minimize the error between the estimated channel matrix $\hat{\boldsymbol{H}}$ and the actual channel matrix $\boldsymbol{H}$, where $\hat{\boldsymbol{H}}$ is estimated from the received signal $\boldsymbol{B}$. We introduce quantization thresholds as

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prior conditions for optimizing the neural network in CGAN, and recover the channel matrix $\hat{\boldsymbol{H}}$ from highly quantized received signal $\boldsymbol{B}$ and known pilot sequences $\boldsymbol{\Phi}$.

III. The Proposed Channel Estimation Method

A. CGAN Structure And Objective Function

The traditional GAN is a method based on adversarial training for training generative models. Its fundamental idea is to train the generator and the discriminator by mutually adversarial means. The learning objective of the generator G is to generate highly realistic data samples from random noise, while the discriminator D needs to learn to distinguish between generated data samples and real data. However, the learning process of the generator may face challenges of instability and randomness. Therefore, researchers have proposed CGAN, which introduces a conditional variable and learns the mapping from conditional input to actual data.

In the context of channel estimation, numerous researchers have investigated the impact of quantization thresholds on channel estimation. In scenarios involving low-bit quantization and low signal-to-noise ratio, the optimal quantization threshold is approximately equal to zero. Many scholars simplify the handling of quantization thresholds during channel estimation by typically setting the threshold size based on the midpoint of the input signal in low-bit situations. However, in many cases, the quantization threshold cannot be handled so simply during the quantization process. Rather, there should exist an optimal quantization threshold to minimize the impact of the quantization error on the channel estimation accuracy.

Traditional algorithms use iterative methods to approximate quantization thresholds, which may encounter convergence issues as the volume of data significantly increases. The quantization threshold, being an uncontrollable unknown parameter, cannot be directly optimized. Recognizing this challenge, in reference [27], the optimal quantization threshold is derived as a function of unknown channel parameters through mathematical formulations. Based on this observation, it is evident that the optimal quantization threshold exhibits a nonlinear relationship with the parameters of the channel matrix. Therefore, leveraging the black-box capability of neural networks, this nonlinear relationship can be effectively approximated. Incorporating the quantization threshold as one of the input conditional information can enhance the input of the CGAN. By harnessing the powerful error-correction ability of the discriminator in CGAN, the input quantization threshold can be made close to the optimal quantization threshold, enabling adaptive quantization and improving the accuracy of channel estimation.

To implement an adaptive quantization scheme based on CGAN to improve channel estimation accuracy, it is necessary to establish the mapping relationship between quantization thresholds and channel matrices. This mapping relationship can be represented as

$$\boldsymbol{\Gamma} : \{\boldsymbol{T}\,\boldsymbol{B}\,\boldsymbol{\Phi}\} \to \{\mathbf{H}\}. \tag{9}$$

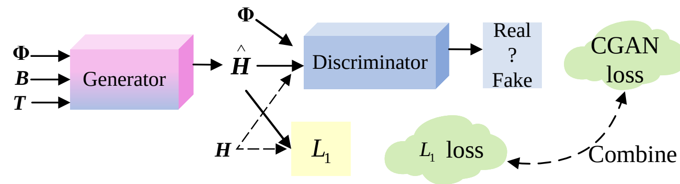

Fig. 2. CGAN structure diagram.

The optimal quantitative threshold [27] is given by the following equation:

$$\tau_n^\star = \boldsymbol{a}_n^T \boldsymbol{h}, \quad \forall n \in \{1, \ldots, N\} \tag{10}$$

After derivation, the optimal solution has been studied in [31] and is given as:

$$XX^H = (P/K)I_K \tag{11}$$

The above results show that when the threshold is optimally selected, the pilot sequences associated with different users should be orthogonal to each other in order to minimize channel estimation error.

In this paper, we utilize CGAN to learn the mapping relationship from the received signal $\boldsymbol{B}$, pilot sequence $\boldsymbol{\Phi}$, and quantization threshold sequence $\boldsymbol{T}$ to the actual channel matrix H. We treat the received signal $\boldsymbol{B}$, quantization threshold sequence $\boldsymbol{T}$, pilot sequence $\boldsymbol{\Phi}$, and channel matrix H as dual-channel images of dimensions $M \times v \times 2$, $M \times v \times 2$,$K \times v \times 2$ and $M \times K \times 2$, respectively, where the two channels of the images represent the real and imaginary parts of complex matrices.

Fig. 2 shows the CGAN structure, including the generator and the discriminator. The quantized received signal $\boldsymbol{B}$ and the pilot sequence $\boldsymbol{\Phi}$ together form the conditional input from which the generator estimates the channel matrix H. The discriminator distinguishes between a given input and labels it as either a true label “1” or a false label “0”. some outliers, which reduces the negative effects of image blurring and makes the resulting image look more realistic at the pixel level. Therefore, to achieve this optimization, we use CGAN loss, denoted as

$$\begin{aligned}\mathscr{L}_{\text{CGAN}}\left(G_\psi, D_\theta, \boldsymbol{B}, \boldsymbol{H}, \boldsymbol{\Phi}, \boldsymbol{T}\right) = E\left[\log D_\theta(\boldsymbol{H}, \boldsymbol{\Phi})\right] + \\ E\left[\log\left(1 - D_\theta\left(G_\psi(\boldsymbol{B}, \boldsymbol{\Phi}, \boldsymbol{T})\right)\right)\right]\end{aligned} \tag{12}$$

where G_ψ represents the generator parameterized by ψ, aimed at synthesizing the channel matrix $\hat{\boldsymbol{H}}$. D_θ is the discriminator parameterized by θ, aimed at distinguishing the generated channel matrix $\hat{\boldsymbol{H}}$ from the real channel $\boldsymbol{H}$. Therefore, we minimize the GAN loss for the generator G_ψ to counteract the loss of the discriminator D_θ, with the objective of maximizing it.

$$\min_\psi \max_\theta \mathscr{L}_{\text{CGAN}}\left(G_\psi, D_\theta, \boldsymbol{B}, \boldsymbol{H}, \boldsymbol{\Phi}, \boldsymbol{T}\right), \tag{13}$$

To ensure that the generator is optimized in the right direction, we augmented the GAN loss with the previously mentioned L_1 loss, which can be expressed as

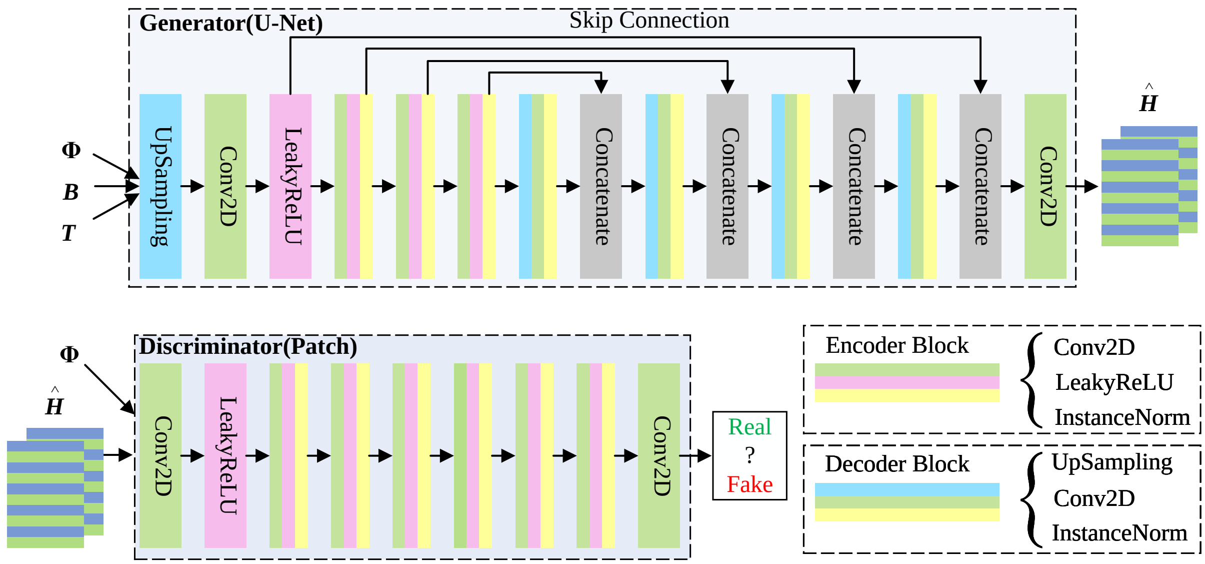

Fig. 3. AQ-CGAN architecture.

$$\mathscr{L}_1 = E\left[\|\boldsymbol{H} - G_\psi(\boldsymbol{B}, \boldsymbol{\Phi}, \boldsymbol{T})\|\right] \tag{14}$$

$$\min_\psi \max_\theta \mathscr{L}_{\text{CGAN}}\left(G_\psi, D_\theta, \boldsymbol{B}, \boldsymbol{H}, \boldsymbol{\Phi}, \boldsymbol{T}\right) + \lambda \mathscr{L}_1, \tag{15}$$

where λ is the weighting factor of the L_1 loss. Once the generator is trained well, channel estimation can be performed based on the new input received signal $\boldsymbol{B}$, pilot sequence $\boldsymbol{\Phi}$, and quantization threshold $\boldsymbol{T}$. The gradient update for parameters ψ and θ can be expressed as

$$\nabla_\theta \frac{1}{m} \sum_{i=1}^{m} \left[\log\left(D_\theta(\boldsymbol{H}, \boldsymbol{\Phi})\right) + \log\left(1 - D_\theta\left(G_\psi(\boldsymbol{B}, \boldsymbol{\Phi}, \boldsymbol{T})\right)\right)\right]. \tag{16}$$

$$\nabla_\psi \frac{1}{m} \sum_{i=1}^{m} \left[\log\left(1 - D_\theta\left(G_\psi(\boldsymbol{B}, \boldsymbol{\Phi}, \boldsymbol{T})\right)\right)\right] \tag{17}$$

where m represents the size of the batch sample in the dataset.

B. Channel Estimation Based on AQ-CGAN

As illustrated in Fig. 3, the generator employs a U-Net architecture [32]. U-Net is a fully convolutional structure designed for image processing. In AQ-CGAN, the generator first utilizes an upsampling layer and a convolutional layer to rescale the input to the same size as $\boldsymbol{H}$. The output of the resizing network is a transformation tracking that can be represented as

$$\begin{aligned} \hat{\mathbf{H}} &= \text{ResizeNet}(\mathbf{B}, \mathbf{T}, \boldsymbol{\Phi}, \boldsymbol{\Omega}) \\ &= Conv(ConvTrans(ConvTrans(\mathbf{B}, \mathbf{T}, \boldsymbol{\Phi}))), \end{aligned} \tag{18}$$

where $Conv(.)$ and $ConvTrans(.)$ represent convolution and deconvolution operation respectively. $\boldsymbol{\Omega} \triangleq \left\{\boldsymbol{w}^{(i)}, \boldsymbol{b}^{(i)}\right\}_{i=1}^{I-1}$ is the network parameters to be trained. Through the resizing network, the dimensions of the input signal can be adjusted to match the dimensions of the channel matrix.

The U-Net architecture comprises three encoders and three decoders. The encoders are composed of convolutional layers, instance normalization layers, and LeakyReLU activation layers, while the decoders consist of upsampling layers, convolutional layers, and instance normalization layers. The introduction of convolutional layers aims to effectively extract image features through their shared weight structure and translation-invariant features. In a convolutional layer, each neuron is connected only to a subset of neighboring neurons from the previous layer, forming the neuron's receptive field, and the same set of weights is shared among neurons throughout the layer. Inspired by the concept of convolution, the encoding process can be expressed using convolutional transformations. Instance normalization is chosen over batch normalization to enhance the convergence efficiency of the generative model. Each convolutional layer is configured with 128 filters of size 4×4 as parameters. The instance normalization layer and ReLU activation layer do not alter the data dimensions. The data volume after the convolution operation is represented as follows

$$G_{out} = \frac{G_{in} + 2p - k}{s}, \tag{19}$$

where G_{out} and G_{in} are the size of output and input respectively. p, s , and k are the size of padding, strides, and kernel size respectively.

The U-Net network architecture introduces skip connections, allowing feature maps from the encoder and decoder modules to be combined using channel connections via skip connections. Compared to traditional encoder-decoder structures, U-Net has this unique advantage. This method enables features extracted by downsampling layers to be directly transmitted to upsampling layers, thereby preserving pixel-level details at different scales and resolutions. Therefore,

Algorithm 1: **AQ-CGAN training process.**

Require: α, the learning rate. c, the batch size. I, the sample size. $\psi_0 \sim N(0, 0.01)$, initialization parameter of the Generator. $\theta_0 \sim N(0, 0.01)$, initialization parameter of the discriminator. N, the number of iterations.

Initialization: $\alpha = 0.0001, c = 100, i = 0, n = 0$.

Training:

1: **For** number of training iterations do
2: **For** i steps do
3: Small batches of samples are obtained from pilot signals, quantized signals, and quantization thresholds. $\{\boldsymbol{\varphi_1}, \ldots, \boldsymbol{\varphi_m}\}$, $\{\boldsymbol{b_1}, \ldots, \boldsymbol{b_m}\}$, $\{\boldsymbol{\tau_1}, \ldots, \boldsymbol{\tau_m}\}$.
4: The samples are inputted into the generator to obtain the generated channel estimation values $\hat{\boldsymbol{H}}$.
5: Update the discriminator by boosting its stochastic gradient through equation (14).
6: Gradient updates optimize the parameters of the discriminator:

$$\boldsymbol{\theta} + \alpha RMSProp\left(\boldsymbol{\theta}, D_\theta\right) \rightarrow \boldsymbol{\theta}$$

7: Update the generator by reducing its stochastic gradient according to equation (15).
8: Gradient updates optimize the parameters of the discriminator:

$$\boldsymbol{\psi} - \alpha Adam\left(\boldsymbol{\psi}, G_\psi\right) \rightarrow \boldsymbol{\psi}$$

9: **End for**
10: **End for**

U-Net exhibits significant effectiveness in preserving details while retaining information at different scales. To avoid mode collapse issues, the generator learns to generate samples by normalizing the output values to the range of $[-1, 1]$ using the hyperbolic tangent activation function, preventing the generation of overly similar samples that could deceive the discriminator.

The discriminator is a commonly used convolutional neural network. The specific structure is shown in Fig. 3, where the patch architecture in the discriminator can enhance the recovery capability of specific detailed information [21]. Unlike conventional discriminators, in this patch architecture, the receptive field can be mapped from the input data. To achieve this, the front part of the discriminator includes a convolutional layer, a LeakyReLU activation layer, and four convolutional layers, all with 512 filters of size 4×4. In the final layer, the fully connected layer is replaced with a convolutional layer to obtain the receptive field, and then the responses of all receptive fields are averaged to obtain the final output. For detailed algorithmic procedure, refer to algorithm 1.

IV. SIMULATION ANALYSIS

A. Dataset Preparation for AQ-CGAN

First, the data for simulation is generated using DeepMIMO. The ray tracing scenario depicts an indoor massive MIMO environment within a room measuring $10\ \mathrm{m} \times 10\ \mathrm{m}$, containing two tables. Antennas are uniformly distributed over a portion of the ceiling, positioned $2.5\ \mathrm{m}$ above the ground. Thirty-two random users are distributed within a portion of the room, all at a height of $1\ \mathrm{m}$. Regarding the propagation model, each channel path can undergo a maximum of two reflections before reaching the receiver. The BS antenna channel dataset is generated utilizing DeepMIMO simulation, where the number of antennas at the BS side is set to $M = 64$ and the number of users remains fixed at $K = 32$. The length of the pilot sequence is 8. The acquisition of quantization thresholds is based on literature [27], which employs statistical knowledge of $\boldsymbol{H}$ to randomly generate a set of distinct thresholds with the hope that some of these thresholds approximate the unknown optimal threshold. It is assumed that each entry of $\boldsymbol{H}$ follows a Gaussian distribution with zero mean and variance of σ^2. While different entries of $\boldsymbol{H}$ may have varying variances, corresponding to different users, for simplicity, it is assumed that all entries have the same variance. Pilot signals are then transmitted over millimeter-wave at a frequency of 28.9 GHz, with Gaussian white noise added to the channel. At the receiver end, CSI is estimated based on the received signal and known pilot signals. The estimated information serves as labels, while the received signals are utilized as inputs to train the network. System parameters are listed in TABLE I.

TABLE I: SIMULATION PARAMETERS

Simulation Parameters	Value
The Number of base station antennas	64
The Number of users	32
The number of antennas per user	1
Length of the pilots	8
Batch size	100

Based on the channel matrix dataset and pilot sequences, datasets corresponding to received signals are generated using 1-bit, 2-bit, and 3-bit quantization methods, with Gaussian white noise added to the received signals spanning signal-to-noise ratios from 0 dB to 25 dB. All datasets are partitioned into training, testing, and validation sets in proportions of $50\%, 40\%$, and 10%, respectively. To expedite convergence speed of the proposed network model under low signal-to-noise ratios while maintaining stability under high signal-to-noise ratios, the generator employs the Adam optimization algorithm, and the discriminator utilizes the RMSProp optimization algorithm, with learning rates set at 2×10^{-4} and 2×10^{-5}, respectively. The input and output coefficients are constrained to known values to improve convergence. To achieve this, normalization is applied to the inputs, multiplied by a scaling factor to increase the range of channel coefficients while maintaining the statistical distribution. This preprocessing method helps to improve the training efficiency of the model. The acquisition of quantization thresholds is based on [27], which suggests randomly generating a set of distinct thresholds based on statistical knowledge of $\boldsymbol{H}$, that some of these thresholds will approximate the unknown optimal thresholds. To facilitate our analysis, we first convert (5) into a real-valued form as follows

$$\widetilde{\boldsymbol{Y}} = \widetilde{\boldsymbol{A}}\widetilde{\boldsymbol{H}} + \widetilde{\boldsymbol{W}} \tag{20}$$

where

$$\begin{aligned} \widetilde{\boldsymbol{Y}} &\triangleq [\Re(\boldsymbol{Y})\Im(\boldsymbol{Y})]^T \\ \widetilde{\boldsymbol{H}} &\triangleq [\Re(\boldsymbol{H})\Im(\boldsymbol{H})]^T \\ \widetilde{\boldsymbol{W}} &\triangleq [\Re(\boldsymbol{W})\Im(\boldsymbol{W})]^T \end{aligned} \tag{21}$$

and

$$\widetilde{\boldsymbol{A}} \triangleq \begin{bmatrix} \Re(\boldsymbol{\Phi}) & \Im(\boldsymbol{\Phi}) \\ -\Im(\boldsymbol{\Phi}) & \Re(\boldsymbol{\Phi}) \end{bmatrix}^T \tag{22}$$

Vectorizing the real-valued matrix $\tilde{\boldsymbol{Y}}$, the received signal can be expressed as a real-valued vector form as

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{h} + \boldsymbol{w} \tag{23}$$

where $\boldsymbol{y} \triangleq \mathrm{vec}(\widetilde{\boldsymbol{Y}}), \boldsymbol{A} \triangleq \boldsymbol{I}_M \otimes \widetilde{\boldsymbol{A}}, \boldsymbol{h} \triangleq \mathrm{vec}(\widetilde{\boldsymbol{H}})$, and $\boldsymbol{w} \triangleq \mathrm{vec}(\widetilde{\boldsymbol{W}})$. Here $\otimes$ stands for the Kronecker product. It can be easily verified $\boldsymbol{y} \in \mathbb{R}^{2Mv}, \boldsymbol{A} \in \mathbb{R}^{2Mv \times 2MK}$, and $\boldsymbol{h} \in \mathbb{R}^{2MK}$

Suppose each entry of $\boldsymbol{h}$ follows a Gaussian distribution with zero mean and variance σ_h^2. Note that different entries of $\boldsymbol{h}$ may have different variances due to that they may correspond to different users. Nevertheless, we assume the same variance for all entries for simplicity. We randomly generate N different realizations of $\boldsymbol{h}$, denoted as $\left\{\tilde{\boldsymbol{h}}_n\right\}$, following this known distribution.

$$\tau_n = \boldsymbol{a}_n^T \tilde{\boldsymbol{h}}_n, \quad \forall n \in \{1, \ldots, N\} \tag{24}$$

where an represents the n-th row of $\boldsymbol{A}$, $N \triangleq 2Mv$. The quantization threshold sequence is obtained according to the above conditions.

We can generate channel matrices with different accuracies by randomly generating different thresholds as inputs to the generator, and then put the generated channel matrices into the discriminator to improve the channel estimation accuracy. During the training process of CGAN network, the discriminator can filter the channel estimation results close to the optimal quantisation threshold. The optimal quantisation threshold cannot simply be treated as 0 under different SNRs, so we need to take this factor into account in our proposed method to approximate the optimal quantisation threshold through the DL method to improve the channel estimation accuracy.

B. Complexity Analysis of The Proposed Algorithm

For a more quantitative analysis, consider the number of multiply-accumulate operations (MACs) of the DNN. This can easily be seen to be proportional to the number of channels, which is simply the number of receive antennas M for a multi-user setup. As shown in TABLE II, we compared several commonly used algorithms. In our scheme, we use the channel simulation values generated according to specific channel scenarios to train the neural network offline. In the case of offline learning, computational complexity is not much concerned, because the time required is usually not strictly limited. Complexity analysis of the channel estimation algorithms used. The complexity analysis of DNN-based channel estimation algorithms takes into account not only the dimensionality of the input data volume, but also the number of layers of the network l. Therefore, the complexity of the DNN method can be expressed as $\mathcal{O}(M^2Kl)$. Considering the AMP algorithm, when the sparsity of the channel vector is set to N, the computational complexity in the atom selection step is about $\mathcal{O}(NMK)$,and the computational complexity of the K-th iteration is at least $\mathcal{O}(N^3)$. Therefore, ignoring trivial operations,the computational complexity can be calculated as $\mathcal{O}(NMK) + \mathcal{O}(N^3M)$. In addition, since both the LAMP network and the GM-LAMP network are constructed based on the AMP algorithm, the computational complexity of the AMP algorithm, the LAMP network, and the GM-LAMP network is roughly the same value equal to $\mathcal{O}(TMK)$. T represents the number of layers in the iteration of the neural network for the LAMP class of methods.

TABLE II: COMPUTATIONAL COMPLEXITY OF SCHEME

Operation	Complexity
DNN	$\mathcal{O}(M^2Kl)$
LAMP	$\mathcal{O}(TMK)$
GM-LAMP	$\mathcal{O}(TMK)$
CGAN	$\mathcal{O}\left(Mk^2\sum_{l=1}^{L} n_{l-1}n_l\right)$
AQ-CGAN	$\mathcal{O}\left((MK+Mv+Kv)\,k^2\sum_{l=1}^{L} n_{l-1}n_l\right)$

The CGAN based channel estimation method is trained to obtain a guide frequency signal and channel estimator that can be effectively used in the online channel estimation process. The complexity of CGAN network is $\mathcal{O}\left(Mk^2\sum_{l=1}^{L} n_{l-1}n_l\right)$, where k is the side length of the convolutional filters, n_{l-1} and n_l denote the numbers of input and output feature maps of the l-th convolutional layer, $1 \leqslant l \leqslant L$. The complexity of channel estimation based on AQ-CGAN comes from full connection network and convolutional network. Its complexity is $\mathcal{O}\left((MK+Mv+Kv)\,k^2\sum_{l=1}^{L} n_{l-1}n_l\right)$.

C. Simulation Results of AQ-CGAN

In this paper, the normalized mean square error (NMSE) is employed as the evaluation metric to assess the disparity between the estimated channel matrix $\hat{\boldsymbol{H}}$ and the true channel matrix $\boldsymbol{H}$, thereby evaluating the performance of channel estimation. Its expression is represented as

$$\mathrm{NMSE} = \mathbb{E}\left[\frac{\left\|\mathbf{H}-\hat{\mathbf{H}}\right\|_2^2}{\|\mathbf{H}\|_2^2}\right]. \tag{25}$$

The experiment was conducted in Python 3.8 environment, Keras 2.4.2, PyCharm Community Edition, with Tensorflow as the backend, on a computer with core i5-6300HQ CPU (2.3GHz). We have established a dataset consisting of 20,000 instances. Through multiple simulations, set the batch_size to

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100. When the batch_size is small, the accuracy rate will be higher, and the learning rate can also be increased to allow the network to converge quickly.

The proposed AQ-CGAN algorithm is compared against several other algorithms under the 1-bit scenario, including the LAMP algorithm, general DNN channel estimation algorithm, GM-LAMP algorithm, and CGAN algorithm. Under the 2-bit and 3-bit scenarios, the main comparison is conducted between the GM-LAMP algorithm, CGAN algorithm, and AQ-CGAN algorithm in terms of channel estimation performance.

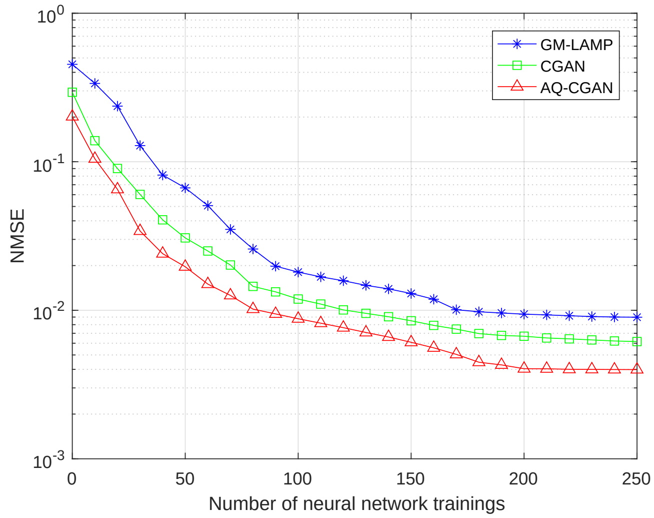

Fig. 4. Convergence of different algorithms for the 1 bit ADC case.

Fig. 4 gives the convergence performance evaluation at a quantization accuracy of 1bit, demonstrating the NMSE of channel estimation versus the number of training sessions. The convergence performance of AQ-CGAN, CGAN and GM-LAMP algorithms are also compared. Due to the introduction of conditional information guide frequency sequence and quantisation threshold in AQ-CGAN algorithm to modify the optimisation direction of the network, AQ-CGAN algorithm converges faster than the other two algorithms as the number of training times increases compared to CGAN algorithm and GM-LAMP algorithm. However, when the number of trainings exceeds 100, the speed of convergence of AQ-CGAN starts to slow down. This is due to the fact that in order to continue to improve the channel estimation accuracy, it is necessary to dig deeper into the mapping relationship between the guide frequency, the quantisation threshold, the quantised signal and the channel matrix. The NMSE performance of the GM-LAMP and CGAN networks stabilises when the number of trainings reaches 170. The NMSE performance of AQ-CGAN stabilises when the number of training times reaches 200. This is due to the fact that the proposed method is more capable of mining more detailed information of the data.

The performance of different channel estimation algorithms for 1bit quantisation scenario and different signal-to-noise ratios is investigated in Fig. 5. Compared to AQ-CGAN other algorithms use conventional quantisation. From the figure, it can be seen that the AQ-CGAN algorithm proposed in this

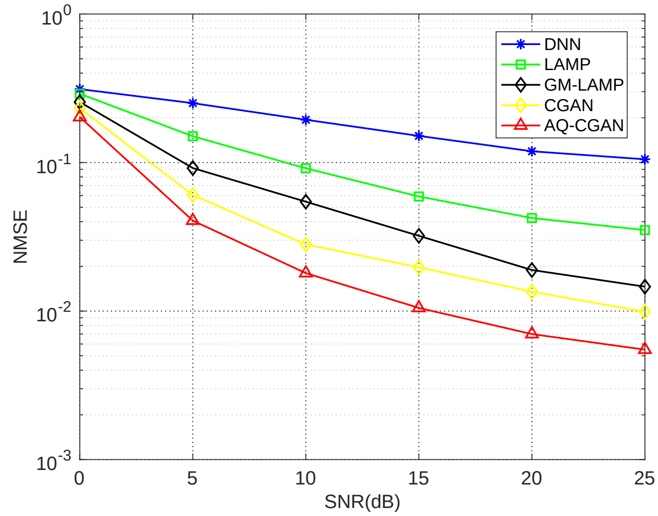

Fig. 5. Under the 1-bit scenario, the NMSE performance of AQ-CGAN and other algorithms at different SNRs.

paper does not have a significant gap with the other compared algorithms at low signal to noise ratios. This is because the value of the optimal quantisation threshold is close to 0 when at low SNR and the other compared algorithms also treat the quantisation threshold as 0. Therefore, there is no significant difference in the performance of these algorithms in this case.

However, as the SNR increases, the LAMP algorithm outperforms the normal DNN algorithm because the LAMP algorithm has a larger training overhead than the DNN, optimises the parameters of the neural network by expanding the iterative algorithm into one layer in the neural network, and sacrifices the complexity of the neural network in exchange for an improvement in the accuracy of the channel estimation. The GM-LAMP algorithm utilizes a new threshold shrinkage function that outperforms the LAMP algorithm in terms of performance. Unlike the fitting functions of DNN, LAMP and GM-LAMP, the CGAN algorithm improves the NMSE performance of the channel estimation by exploiting the a priori information to improve the accuracy of the channel estimates. AQ-CGAN further improves the channel estimation accuracy by introducing a quantisation threshold on the basis of CGAN. When the SNR is 25dB, the channel estimation accuracy of AQ-CGAN is improved by about 0.28% over CGAN and 8.1% over normal DNN channel estimation accuracy. It can be observed through the curve trend that the AQ-CGAN algorithm outperforms the above four algorithms and shows better channel estimation performance.

Fig. 6 compares the NMSE performance of AQ-CGAN, CGAN, and GM-LAMP algorithms under different SNRs when the ADC quantizer is set to 2 bits. From the graph, it is evident that the NMSE performance of the AQ-CGAN algorithm surpasses that of CGAN and GM-LAMP algorithms. Moreover, when the ADC quantizer adopts 2-bit quantization, the loss of signal amplitude information is less severe compared to 1-bit quantization, thereby resulting in less quantization error. Consequently, as the SNR increases, the channel estimation errors of these three algorithms are

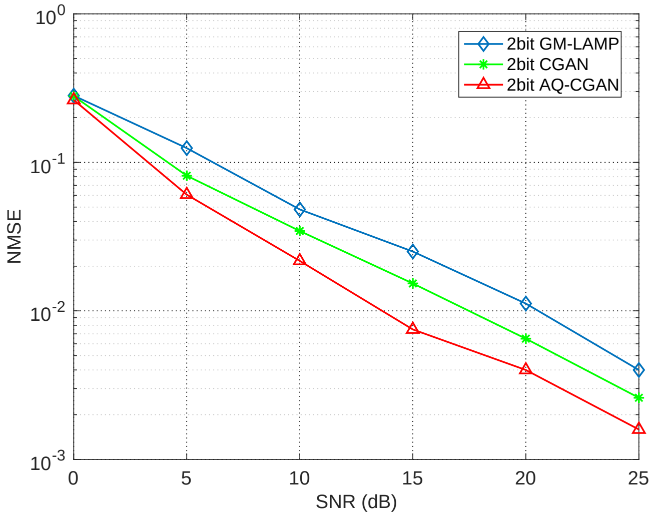

Fig. 6. Under the 2-bit scenario, the NMSE performance of AQ-CGAN and other algorithms at different SNRs.

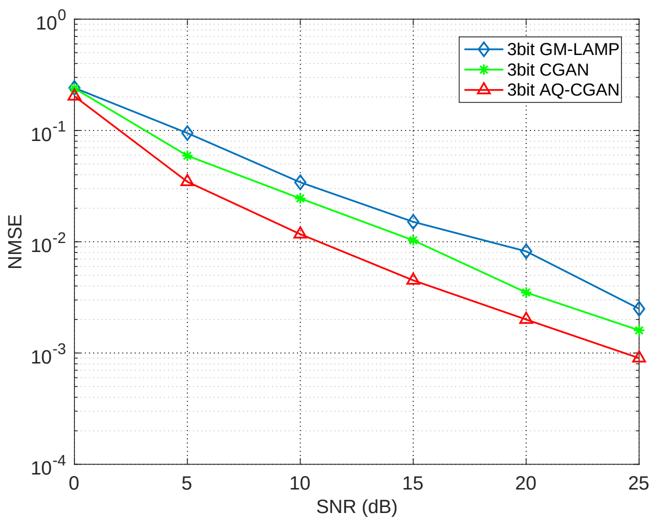

Fig. 7. Under the 3-bit scenario, the NMSE performance of AQ-CGAN and other algorithms at different SNRs.

significantly lower than those under 1-bit quantization. A direct comparison between AQ-CGAN and CGAN algorithms reveals that under low SNR conditions, where the optimal quantization threshold is close to zero, the difference in NMSE performance between the two is not pronounced. However, as the SNR increases, the accuracy gap in channel estimation between the two algorithms widens. This observation further underscores the significant performance improvement of the proposed algorithm compared to the benchmark algorithms.

Fig. 7 presents a comparative analysis of the NMSE performance of AQ-CGAN, CGAN, and GM-LAMP algorithms under a 3-bit ADC quantization precision. To validate the performance superiority of the proposed AQ-CGAN algorithm, NMSE performance comparisons were conducted among AQ-CGAN, CGAN, and GM-LAMP algorithms across three quantization scenarios: 1-bit, 2-bit, and 3-bit. Through the comparisons illustrated in Fig. 5, Fig. 6, and Fig. 7, it is evident that the proposed AQ-CGAN algorithm exhibits performance advantages across all three quantization scenarios. Additionally, the accuracy of channel estimation is significantly improved compared to the CGAN algorithm.

V. CONCLUSION

The use of low-resolution ADCs instead of high-resolution ADCs can significantly reduce the power consumption and hardware cost of millimeter-wave large-scale multiple-input multiple-output systems. However, conventional channel estimation methods cannot handle the introduced nonlinear effects. In addition, previous deep learning-based methods do not consider the effect of quantization threshold on channel estimation performance. In this paper, quantization threshold is introduced to improve the input of CGAN, and an AQ-CGAN algorithm is proposed to realize adaptive quantization by establishing a mapping between the quantization threshold and the channel matrix to reduce the nonlinear error induced by low-resolution ADC. Simulation results show that the proposed AQ-CGAN algorithm significantly improves the estimation accuracy in all three quantization accuracy cases and achieves better channel estimation performance. Future work will focus on the adaptive quantization algorithm under mixed-precision ADCs.

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